

Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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Abstract—This document presents the mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM).

I. INTRODUCTION

A. Problem in One Sentence

After Scout UAVs detect survivors, the Command Center must decide at each event time t which available Responder UAVs should execute which feasible tours—ordered sequences $\pi_j = (i_1, \dots, i_k)$ —to maximize expected lifesaving value under battery, payload-item, and 3D reachability constraints, before ground rescue arrives.

II. SYSTEM ENTITIES

A. Survivors

Detected tasks i , each with location $p_i \in \mathbb{R}^3$, detection time t_i^{det} , confidence $c_i \in [0, 1]$, severity $\sigma_i \in [0, 1]$, and demand vector $r_i^q \in \mathbb{Z}_{\geq 0}$ for $q \in \{\text{med}, \text{water}, \text{food}\}$.

B. Responder UAVs

Platforms j that execute tours, each with current position p_j , base $p_j^{\text{base}} \in \mathbb{R}^3$, battery $B_j(t) \in [0, B_{\max}]$, item capacity K , inventory I_j^q , payload $m_j = \sum_q I_j^q w_q$, and status $\in \{\text{Idle}, \text{EnRoute}\}$.

C. Command Center

Central decision unit that maintains global state $X(t)$, queries the routing oracle for each candidate tour, solves the assignment optimization, and dispatches tours π_j .

D. Environment

Disaster area $\mathcal{E} = (V, O)$ with $V \subset \mathbb{R}^3$ free space and $O \subset \mathbb{R}^3$ static obstacles (buildings, rubble), $V \cap O = \emptyset$; hazard $R(p)$ is deferred to $R = 0$ in V1.

III. SYSTEM STATE AT TIME t

A. Survivor Set

$\mathcal{S}(t) = \{S_i\}$, $\mathcal{Q}_{\text{high}}(t) = \{i \in \mathcal{S}(t) \mid a_i = 0 \wedge c_i \geq c_{\text{thresh}}\}$ ordered by $V_i(t)$, $\mathcal{Q}_{\text{low}}(t) = \{i \mid a_i = 0 \wedge c_i < c_{\text{thresh}}\}$.

B. UAV Set

$\mathcal{D}(t) = \{D_j\}$, $\mathcal{D}_{\text{avail}}(t) = \{j \mid \text{status}_j = \text{Idle}\}$.

C. Global State

$X(t) = (\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}, \mathcal{Q}_{\text{high}}(t), \mathcal{Q}_{\text{low}}(t))$.

IV. SURVIVOR PRIORITY AND VALUE

A. Urgency

$u_i(t) = \sigma_i e^{-\alpha(t-t_i^{\text{det}})}$, $\alpha = \ln 2 / T_{\text{half}}$, $T_{\text{half}} = 30$ min, $u_i \in [0, 1]$ proportional decay $du/dt = -\alpha u$, half-life, never negative.

B. Service Value

$V_i(t) = \frac{2u_i(t)c_i}{u_i(t)+c_i} \in [0, 1]$ (harmonic mean, $V_i = 0$ if $u_i = 0$ or $c_i = 0$) unified expected value; $c_i \geq c_{\text{thresh}}$ hard eligibility for $\mathcal{Q}_{\text{high}}$.

C. Queue Priority

priority(i) = $\beta V_i(t)$, $\mathcal{Q}_{\text{high}}$ ordered by V_i descending.

V. UAV MISSION AND TOUR

A. Tour Definition

$\pi_j(t) = (i_1, \dots, i_k)$ with $k = \text{len}(\pi_j)$ and $\sum_{i \in \pi_j} \sum_q r_i^q \leq K$ (e.g., $K = 4$). $P_{\pi_j} = P_{p_j \rightarrow p_{i_1}} \circ \dots \circ P_{p_{i_k} \rightarrow p_j^{\text{base}}}$.

B. Example

$K = 4$, $I_j = [\text{med}, \text{water}, \text{food}, \text{med}]$, $\pi_j = (A, C)$ with $\mathcal{R}_A = \{\text{med}\}$, $\mathcal{R}_C = \{\text{water}, \text{food}\}$ carries 3 items to 2 survivors: base(2.8kg) $\rightarrow$ A(10s, 3300J) $\rightarrow$ C(5s, 1500J) $\rightarrow$ base(12s).

VI. ROUTING MODULE

For any path $P = \{p_0, \dots, p_L\} \subset V$, let $T(P) = \sum_k \|p_k - p_{k-1}\| / v_{\max}$ be travel time and $E(P) = \sum_k [e_{\text{travel}}(p_k, m, w_0) \|p_k - p_{k-1}\| + e_{\text{vert}}(m) \max(0, z_k - z_{k-1})]$ be energy, where L is waypoints. A path P is feasible iff $P \subset V$.

VII. TOUR FEASIBILITY

A tour $\pi_j = (i_1, \dots, i_k)$ is feasible iff:

- **Capacity:** $\sum_{i \in \pi_j} \sum_q r_i^q \leq K$,
- **Inventory:** $I_j^{q,(l)} = I_j^{q,(l-1)} - r_{i_l}^q$, $I_j^{q,(l)} \geq 0 \forall q, l$,
- **Battery:** $E(\pi_j) + E_{\text{reserve}} \leq B_j(t)$,
- **Reachability:** $P_l \subset V \forall l$,
- **Confidence:** $c_i \geq c_{\text{thresh}} \forall i \in \pi_j$.

If any fails, Feasible(π_j) = false and $C(\pi_j) = \infty$.

VIII. SYSTEM MODEL	
IX. PROBLEM FORMULATION	
X. CONCLUSION	